Experiment 3: Transistor Biasing

Malachy Crossan & Brandon Saavedra

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2.A Objective

To study the characteristics and the applications of bipolar junction transistors

2.B Pre-lab

2.B.1 Part A

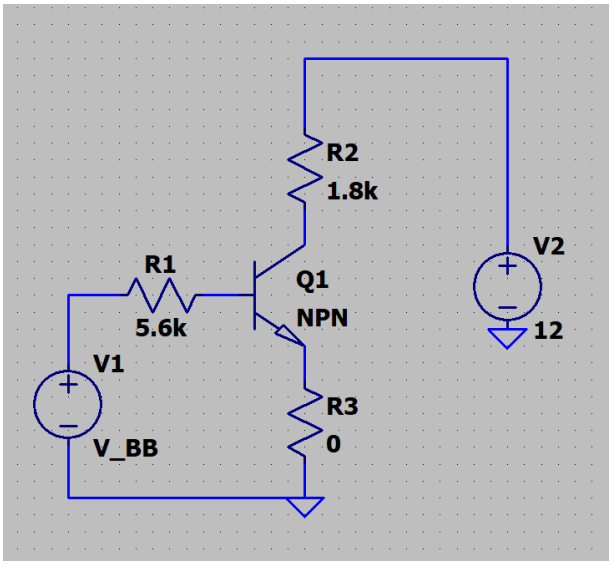

Figure 1: Circuit 1.b

2.B.2 Part B

$$V_{\mathbb{C}} = 12V$$
$$R_C = 1.8k\Omega \; R_B = 5.6k\Omega \; R_E = 0\Omega$$
$$\beta = 220 \text{ and } V_{\text{BE}} = 0.7V$$
$$I_C = 2mA$$

Eq 1

$$\frac{I_C}{\beta} = I_B$$
$$\frac{2mA}{220} = 9.09\mu A$$

Eq 2

$$V_{\text{BE}} = 0.7V$$
$$\frac{V_{\text{BB}} - V_{\text{BE}}}{R_{\text{BB}}} = I_B$$
$$\frac{V_{\text{BB}} - 0.7}{5.6k\Omega} = 9.09\mu A$$
$$V_{\text{BB}} = 750mV$$

Eq 3

2.B.3 Part C